PHC213 · Fall 2026

Pharmaceutical Organic Chemistry 3

Question bank — 137 multiple-choice questions across 6 lectures. Answer key with explanations at the end.

Lecture 1 — Aldehydes and ketones: names, properties, preparation

1. Oxidizing a primary alcohol with PCC gives:
A. An aldehyde
B. A carboxylic acid
C. A ketone
D. An alkene

2. In the Rosenmund reduction, $BaSO_4$ is used to:
A. Speed up the reaction
B. Poison the Pd catalyst so the aldehyde is not reduced further
C. Provide acidity
D. Remove water

3. Friedel–Crafts acylation of benzene with $RCOCl/AlCl_3$ prepares:
A. Aromatic aldehydes
B. Phenols
C. Aromatic ketones
D. Carboxylic acids

4. Which functional group has the highest IUPAC priority?
A. −CHO
B. −COOR
C. −COOH
D. −OH

5. Aldehydes and ketones have lower boiling points than alcohols because they:
A. Are non-polar
B. Are ionic
C. Have lower molecular weight
D. Cannot H-bond with each other

6. The carbonyl carbon in aldehydes and ketones is hybridized:
A. sp
B. sp^3
C. sp^2
D. dsp^2

7. The C=O bond is polar with partial positive charge on:
A. Oxygen
B. Hydrogen
C. Carbon
D. Both equally

8. The IUPAC suffix for aldehydes is:
A. -one
B. -al
C. -ol
D. -oic acid

9. CH_3COCH_3 is named:
A. Propanal
B. Butanone
C. Ethanal
D. Propanone

10. HCHO is:
A. Acetaldehyde
B. Acetone
C. Formaldehyde (methanal)
D. Benzaldehyde

11. Aldehydes and ketones have boiling points:
A. Higher than alcohols of similar mass
B. Lower than alkanes
C. Lower than alcohols but higher than alkanes
D. Same as alkanes

12. Lower aldehydes and ketones are water soluble because:
A. They ionize
B. They react with water fully
C. They are non-polar
D. They form H-bonds with water through the carbonyl oxygen

13. Oxidation of a primary alcohol with PCC gives:
A. Carboxylic acid
B. Aldehyde
C. Ketone
D. Alkene

14. Oxidation of a secondary alcohol gives:
A. Aldehyde
B. Ether
C. Acid
D. Ketone

15. Ozonolysis of an alkene followed by Zn/H_2O gives:
A. Aldehydes and/or ketones
B. Alcohols
C. Acids only
D. Alkanes

16. Hydration of a terminal alkyne ($HgSO_4/H_2SO_4$) gives:
A. An aldehyde
B. A methyl ketone
C. A carboxylic acid
D. An alcohol

17. Hydroboration–oxidation of a terminal alkyne gives:
A. A methyl ketone
B. An alkene
C. An aldehyde
D. An ester

18. Friedel–Crafts acylation of benzene with $CH_3COCl/AlCl_3$ gives:
A. Acetophenone
B. Toluene
C. Benzaldehyde
D. Phenol

19. Rosenmund reduction converts acid chlorides to:
A. Alcohols
B. Alkanes
C. Ketones
D. Aldehydes

20. Gattermann–Koch reaction (CO + HCl, $AlCl_3$/CuCl) on benzene gives:
A. Benzoic acid
B. Acetophenone
C. Benzaldehyde
D. Chlorobenzene

21. Etard reaction (CrO_2Cl_2) converts toluene to:
A. Benzaldehyde
B. Benzoic acid
C. Benzyl alcohol
D. Phenol

22. Reaction of a nitrile with a Grignard reagent then hydrolysis gives:
A. Ketone
B. Aldehyde
C. Primary amine
D. Alcohol

Lecture 2 — Nucleophilic addition, stereochemistry and reduction

23. Which is most reactive toward nucleophilic addition?
A. Acetone
B. Benzaldehyde
C. Acetaldehyde
D. Acetophenone

24. An acetal is formed from an aldehyde and:
A. Two equivalents of alcohol, dry HCl
B. One equivalent of alcohol, base
C. HCN
D. Hydrazine

25. By the Cram & Abdel-Hafez rule, the major product forms when the nucleophile attacks from the side of the:
A. Small group, giving threo
B. Large group, giving erythro
C. Medium group, giving threo
D. Either side equally

26. Reducing a ketone with $NaBH_4$ gives:
A. A primary alcohol
B. A carboxylic acid
C. An alkane
D. A secondary alcohol

27. Which converts C=O directly to CH_2?
A. $NaBH_4$
B. Clemmensen reduction
C. PCC
D. Cannizzaro

28. In a Fischer projection the lowest-priority group is horizontal and $1 \rightarrow 2 \rightarrow 3$ runs clockwise. The configuration is:
A. S
B. R
C. Erythro
D. Cannot tell

29. The characteristic reaction of aldehydes and ketones is:
A. Electrophilic addition
B. Electrophilic substitution
C. Free-radical substitution
D. Nucleophilic addition

30. Aldehydes are generally more reactive than ketones because of:
A. Larger size
B. Aromaticity
C. Higher molecular weight
D. Less steric hindrance and fewer electron-donating alkyl groups

31. Nucleophilic addition to a carbonyl changes carbon hybridization from:
A. sp^3 to sp^2
B. sp^2 to sp^3
C. sp to sp^2
D. Unchanged

32. Attack of a nucleophile on an unsymmetrical ketone (e.g. butanone) creates:
A. A meso compound
B. A racemic mixture (attack from both faces)
C. A single enantiomer
D. No stereocentre

33. HCN adds to carbonyls to give:
A. Imines
B. Cyanohydrins
C. Acetals
D. Hydrates

34. Aldehydes react with excess alcohol in acid to form:
A. Acetals
B. Hemiacetals only
C. Ketals only from aldehydes
D. Esters

35. Acetals are useful as:
A. Oxidants
B. Catalysts
C. Acids
D. Protecting groups for carbonyls (stable in base)

36. Primary amines react with aldehydes/ketones to give:
A. Enamines
B. Imines (Schiff bases)
C. Amides
D. Nitriles

37. Hydroxylamine reacts with carbonyls to form:
A. Hydrazones
B. Imines
C. Semicarbazones
D. Oximes

38. 2,4-Dinitrophenylhydrazine gives with carbonyls:
A. A colourless solution
B. Yellow/orange-red precipitate (2,4-DNP derivative)
C. A silver mirror
D. Blue colour

39. Grignard reagent with formaldehyde, then H_3O^+, gives:
A. Secondary alcohol
B. Primary alcohol
C. Tertiary alcohol
D. Ketone

40. Grignard reagent with a ketone, then H_3O^+, gives:
A. Tertiary alcohol
B. Secondary alcohol
C. Primary alcohol
D. Aldehyde

41. $NaBH_4$ reduces aldehydes to:
A. Primary alcohols
B. Secondary alcohols
C. Alkanes
D. Acids

42. Clemmensen reduction (Zn(Hg)/HCl) converts C=O to:
A. CH–OH
B. CH_2
C. C=C
D. COOH

43. Wolff–Kishner reduction uses:
A. Zn(Hg)/HCl
B. $LiAlH_4$
C. NH_2NH_2/KOH, heat
D. H_2/Pd

44. The Wittig reaction converts a carbonyl into:
A. An alcohol
B. An amine
C. An ester
D. An alkene

45. Tollens' reagent gives a silver mirror with:
A. Ketones
B. Alkanes
C. Ethers
D. Aldehydes

46. Fehling's solution gives a red precipitate of:
A. Ag
B. Cu_2O
C. CuO
D. Fe_2O_3

Lecture 3 — α-Hydrogen reactions and carboxylic acids

47. Which aldehyde undergoes the Cannizzaro reaction?
A. Acetaldehyde
B. Benzaldehyde
C. Propanal
D. Butanal

48. A positive iodoform test needs which group?
A. −CHO
B. CH_3CO-
C. −COOH
D. −OH on a ring

49. The aldol condensation requires:
A. No α-hydrogen
B. An acid catalyst and no water
C. An α-hydrogen and a strong base
D. A Grignard reagent

50. The IUPAC name of $CH_3CH(Cl)CH_2CH_2COOH$ is:
A. 2-chloropentanoic acid
B. 4-chloropentanoic acid
C. 4-chlorobutanoic acid
D. 1-chloropentanoic acid

51. An α-hydrogen is a hydrogen on the carbon:
A. Of the carbonyl itself
B. Adjacent to the carbonyl
C. Two carbons away
D. On oxygen

52. α-Hydrogens are acidic because the conjugate base (enolate) is:
A. Unstable
B. A radical
C. Aromatic
D. Resonance-stabilized with charge on oxygen

53. Keto–enol tautomerism involves:
A. Movement of a proton and a double bond
B. Change in molecular formula
C. Loss of water
D. Radical formation

54. Aldol condensation requires the carbonyl compound to have:
A. No α-H
B. An aromatic ring
C. At least one α-H
D. A halogen

55. The aldol product of acetaldehyde is:
A. Crotonic acid
B. Butanone
C. 3-Hydroxybutanal
D. Ethanol

56. Heating an aldol product gives:
A. An α,β-unsaturated carbonyl
B. An alcohol
C. An acid
D. An ether

57. The Cannizzaro reaction occurs with aldehydes that:
A. Have α-H
B. Are ketones
C. Lack α-H, in concentrated base
D. Are acidic

58. Cannizzaro reaction of benzaldehyde gives:
A. Benzyl alcohol + benzoate
B. Two molecules of benzoic acid
C. Toluene
D. Benzene

59. In a crossed Cannizzaro with HCHO, formaldehyde is:
A. Oxidized to formate
B. Reduced to methanol
C. Unchanged
D. Polymerized

60. The haloform (iodoform) test is positive for:
A. All ketones
B. Methyl ketones and ethanol/2° alcohols $CH_3CH(OH)–$
C. Formaldehyde
D. Benzaldehyde

61. Iodoform has the formula:
A. CHI_3
B. CH_2I_2
C. CI_4
D. CH_3I

62. Claisen–Schmidt condensation is between:
A. Two ketones
B. An aromatic aldehyde (no α-H) and a ketone/aldehyde with α-H
C. Two esters
D. An acid and alcohol

63. The carboxyl group combines:
A. C–O–C
B. C=O and $–NH_2$
C. C=O and –OH
D. C≡N

64. Carboxylic acids have high boiling points because they form:
A. Ionic lattices
B. No H-bonds
C. Covalent polymers
D. Hydrogen-bonded dimers

65. The IUPAC name of CH_3COOH is:
A. Methanoic acid
B. Propanoic acid
C. Ethanoic acid
D. Acetic aldehyde

66. HCOOH is:
A. Acetic acid
B. Formic (methanoic) acid
C. Oxalic acid
D. Benzoic acid

Lecture 4 — Acidity, preparation and reactions of carboxylic acids

67. Which acid is strongest?
A. ICH_2COOH
B. $BrCH_2COOH$
C. FCH_2COOH
D. $ClCH_2COOH$

68. o-Toluic acid is stronger than benzoic acid even though CH_3 donates electrons because of:
A. Hydrogen bonding
B. Hyperconjugation
C. The ortho effect (steric inhibition of resonance)
D. The +M effect

69. Reacting a Grignard reagent with CO_2 gives a carboxylic acid with:
A. One more carbon
B. The same number of carbons
C. One fewer carbon
D. Two more carbons

70. The Hell–Volhard–Zelinsky reaction:
A. Halogenates the α-carbon of acids (X_2/P)
B. Reduces acids to alcohols
C. Makes esters
D. Nitrates benzoic acid

71. Why is diclofenac potassium preferred for acute pain over diclofenac sodium?
A. It is a different drug
B. It has fewer side effects
C. It lasts longer
D. It is more water-soluble, so absorbed faster

72. Which carboxylic acid derivative is the most reactive?
A. Amide
B. Ester
C. Carboxylate
D. Acid chloride

73. Carboxylic acids are more acidic than alcohols because the carboxylate is:
A. Localized
B. Neutral
C. Aromatic
D. Resonance-stabilized (two equivalent O)

74. Electron-withdrawing groups on the α-carbon:
A. Decrease acidity
B. No effect
C. Increase acidity
D. Make it basic

75. Which is the strongest acid?
A. CH_3COOH
B. $ClCH_2COOH$
C. $Cl_2CHCOOH$
D. Cl_3CCOOH

76. Which is more acidic: FCH_2COOH or $ClCH_2COOH$?
A. FCH_2COOH
B. $ClCH_2COOH$
C. Equal
D. Neither is acidic

77. The inductive effect decreases with:
A. Distance from the carboxyl group
B. Number of halogens
C. Electronegativity
D. Temperature

78. Formic acid is stronger than acetic acid because:
A. H is electron-withdrawing
B. Formic acid is aromatic
C. CH_3 is electron-donating (+I), destabilizing acetate
D. Formic acid has two COOH

79. p-Nitrobenzoic acid is ___ than benzoic acid.
A. Less acidic
B. Not acidic
C. Equally acidic
D. More acidic

80. p-Methoxybenzoic acid is ___ than benzoic acid.
A. More acidic
B. Less acidic
C. Equal
D. Basic

81. Ortho-substituted benzoic acids are usually stronger acids due to the:
A. Ortho effect (steric + electronic)
B. Para effect
C. Meta effect
D. No effect

82. Carboxylic acids react with $NaHCO_3$ to release:
A. H_2
B. CO_2
C. O_2
D. Cl_2

83. Oxidation of toluene with hot $KMnO_4$ gives:
A. Benzaldehyde
B. Benzyl alcohol
C. Benzoic acid
D. Phenol

84. Hydrolysis of a nitrile (R–C≡N) in acid gives:
A. Amine
B. Carboxylic acid
C. Aldehyde
D. Ketone

85. Grignard reagent + CO_2 then H_3O^+ gives:
A. Alcohol
B. Ketone
C. Carboxylic acid with one extra carbon
D. Aldehyde

86. Fischer esterification is reaction of a carboxylic acid with:
A. An amine
B. An alcohol in acid catalysis
C. $SOCl_2$
D. NaOH

87. $SOCl_2$ converts carboxylic acids to:
A. Esters
B. Amides
C. Acid chlorides
D. Anhydrides

88. $LiAlH_4$ reduces carboxylic acids to:
A. Primary alcohols
B. Aldehydes
C. Alkanes
D. Ketones

89. Heating a β-keto acid or malonic acid causes:
A. Esterification
B. Decarboxylation (loss of CO_2)
C. Hydration
D. Polymerization

90. Hell–Volhard–Zelinsky reaction introduces:
A. OH at para position
B. α-Halogen into carboxylic acids (Br_2/P)
C. A nitro group
D. An ester group

Lecture 5 — Conformations and cyclohexane

91. The most stable conformation of cyclohexane is the:
A. Planar form
B. Chair
C. Boat
D. Half-chair

92. The flagpole interaction in boat cyclohexane is between hydrogens on:
A. C1 and C2
B. C1 and C3
C. C2 and C3
D. C1 and C4

93. For ethylene glycol, the most stable conformer is gauche because of:
A. Torsional strain
B. Angle strain
C. Steric repulsion
D. An intramolecular hydrogen bond

94. Conformations are:
A. Isomers requiring bond breaking
B. Structural isomers
C. Different spatial arrangements from rotation about single bonds
D. Enantiomers only

95. In ethane, the more stable conformation is:
A. Eclipsed
B. Both equal
C. Staggered
D. Boat

96. The energy barrier for ethane rotation is about:
A. 0.3 kcal/mol
B. 3 kcal/mol (12 kJ/mol)
C. 30 kcal/mol
D. 300 kcal/mol

97. In butane, the most stable conformation is:
A. Anti
B. Gauche
C. Fully eclipsed
D. Partially eclipsed

98. Gauche butane is less stable than anti because of:
A. Steric strain between methyl groups at 60°
B. Torsional strain only
C. Angle strain
D. Ring strain

99. The most stable conformation of cyclohexane is:
A. Boat
B. Twist-boat
C. Chair
D. Planar

100. Bond angles in chair cyclohexane are about:
A. 90°
B. 109.5°
C. 120°
D. 180°

101. The boat conformation is destabilized by:
A. Hydrogen bonds
B. Angle strain
C. Aromaticity
D. Flagpole interactions and eclipsing

102. Each carbon in chair cyclohexane has:
A. Two axial H
B. One axial and one equatorial H
C. Two equatorial H
D. No H

103. Axial bonds are:
A. Roughly in the ring plane
B. Parallel to the ring's axis (up/down)
C. At 45°
D. Absent

104. Ring flipping converts axial substituents to:
A. Axial
B. Planar
C. Equatorial
D. Eclipsed

105. A bulky substituent prefers the equatorial position to avoid:
A. 1,3-Diaxial interactions
B. Torsional strain
C. Angle strain
D. Resonance

106. tert-Butylcyclohexane exists almost entirely with t-Bu:
A. Axial
B. In boat form
C. Both equally
D. Equatorial

107. In cis-1,2-disubstituted cyclohexane, substituents are:
A. One axial, one equatorial
B. Both equatorial
C. Both axial
D. Planar

108. The more stable isomer of 1,4-dimethylcyclohexane is:
A. cis
B. Both equal
C. trans (diequatorial)
D. Neither

109. In cis-1,3-dimethylcyclohexane the most stable conformer is:
A. Diaxial
B. Diequatorial
C. Axial–equatorial
D. Boat

110. Cyclopropane is highly strained because its bond angle is:
A. 60° vs 109.5° ideal
B. 109.5°
C. 120°
D. 90°

111. Newman projection views a molecule:
A. Along a C–C bond
B. From above the ring
C. As a 2D skeletal formula
D. As a wedge-dash drawing only

Lecture 6 — Nomenclature of heterocycles

112. A saturated 3-membered ring containing one nitrogen is called:
A. Azirine
B. Oxirane
C. Aziridine
D. Azetidine

113. About what share of US FDA-approved drugs contain nitrogen heterocycles?
A. 10%
B. 25%
C. 59%
D. 90%

114. A heterocycle contains in its ring:
A. At least one atom other than carbon
B. Only carbon
C. Only nitrogen
D. Metals only

115. In Hantzsch–Widman nomenclature the prefix for oxygen is:
A. Thia
B. Aza
C. Oxa
D. Phospha

116. When several heteroatoms are present, priority order is:
A. N > O > S
B. S > O > N
C. O > S > N
D. N > S > O

117. The final 'a' of a prefix is dropped before:
A. A consonant
B. Nothing
C. A number
D. A vowel

118. The Hantzsch–Widman stem for a saturated 3-membered ring without N is:
A. -irene
B. -irane
C. -ole
D. -ine

119. The stem for an unsaturated 5-membered ring is:
A. -olane
B. -ole
C. -ine
D. -epine

120. A saturated 5-membered ring with one oxygen is named:
A. Oxolane
B. Oxole
C. Oxirane
D. Oxane

121. Furan in Hantzsch–Widman terms is:
A. Oxolane
B. Oxane
C. Oxole
D. Oxirene

122. Thiophene contains:
A. O
B. S
C. N
D. P

123. Pyrrole is a 5-membered ring with:
A. One O
B. One S
C. Two N
D. One NH

124. Pyridine is:
A. 6-membered aromatic with one N
B. 5-membered with N
C. 6-membered with O
D. Saturated

125. Saturated 6-membered ring with one N is:
A. Pyridine
B. Piperidine
C. Pyrrolidine
D. Morpholine

126. Imidazole has nitrogens at positions:
A. 1,2
B. 1,3
C. 1,4
D. 2,3

127. Pyrazole is:
A. 1,3-Diazole
B. 1,3-Oxazole
C. 1,2-Diazole
D. Thiazole

128. Thiazole contains:
A. O and N at 1,2
B. S and N at 1,3
C. Two S
D. S and O

129. Oxazole versus isoxazole: isoxazole has O and N at:
A. 1,3
B. 1,2
C. 1,4
D. 2,5

130. Pyrimidine has nitrogens at positions:
A. 1,2
B. 1,3
C. 1,4
D. 2,4

131. Pyrazine is:
A. 1,2-Diazine
B. 1,3-Diazine
C. 1,4-Diazine
D. Triazine

132. Indole is benzene fused to:
A. Pyrrole
B. Pyridine
C. Furan
D. Thiophene

133. Quinoline is benzene fused to pyridine with N at position:
A. 1
B. 2
C. 3
D. 4

134. Purine is a fusion of:
A. Two pyridines
B. Pyridine and pyrrole
C. Benzene and furan
D. Pyrimidine and imidazole

135. The 7-membered unsaturated ring stem is:
A. -ine
B. -epine
C. -ocine
D. -ole

136. Numbering in a heterocycle starts at:
A. The heteroatom of highest priority
B. Any carbon
C. The substituent
D. The largest ring

137. The aromaticity of furan, pyrrole and thiophene requires:
A. 6 π electrons including a heteroatom lone pair
B. 4 π electrons
C. 8 π electrons
D. No lone pair

Answer key

Lecture 1

1	**A**	**An aldehyde** — PCC is mild: 1° alcohol stops at the aldehyde; 2° alcohol gives a ketone.
2	**B**	**Poison the Pd catalyst so the aldehyde is not reduced further** — Without poisoning, the aldehyde would be reduced on to the alcohol.
3	**C**	**Aromatic ketones** — It is used for aromatic ketones only.
4	**C**	**−COOH** — COOH > COOR > COCl > $CONH_2$ > CN > CHO > C=O > OH > NH_2.
5	**D**	**Cannot H-bond with each other** — They are polar but have no O–H, so no intermolecular H-bonding.
6	**C**	**sp^2** — Trigonal planar, ~120°.
7	**C**	**Carbon** — O is more electronegative; C is electrophilic.
8	**B**	**-al** — Ketones use -one.
9	**D**	**Propanone** — Common name acetone.
10	**C**	**Formaldehyde (methanal)** — The simplest aldehyde.
11	**C**	**Lower than alcohols but higher than alkanes** — Dipole–dipole but no H-bonding between molecules.
12	**D**	**They form H-bonds with water through the carbonyl oxygen** — Solubility falls as the chain grows.
13	**B**	**Aldehyde** — PCC stops at the aldehyde.
14	**D**	**Ketone** — E.g. with $K_2Cr_2O_7/H^+$.
15	**A**	**Aldehydes and/or ketones** — Reductive work-up stops at carbonyls.
16	**B**	**A methyl ketone** — Markovnikov; via enol tautomerization.
17	**C**	**An aldehyde** — Anti-Markovnikov enol → aldehyde.
18	**A**	**Acetophenone** — An aryl ketone.
19	**D**	**Aldehydes** — H_2/Pd–$BaSO_4$ (poisoned).
20	**C**	**Benzaldehyde** — Formylation of arenes.
21	**A**	**Benzaldehyde** — Stops at the aldehyde.
22	**A**	**Ketone** — Via an imine salt.

Lecture 2

23	**C**	**Acetaldehyde** — Aliphatic aldehyde: least hindered and no resonance stabilization of C=O.
24	**A**	**Two equivalents of alcohol, dry HCl** — Via a hemiacetal; reversible by adding water.
25	**A**	**Small group, giving threo** — Least steric hindrance: small-group side → threo (major).
26	**D**	**A secondary alcohol** — Aldehyde → 1° alcohol; ketone → 2° alcohol.
27	**B**	**Clemmensen reduction** — Clemmensen (Zn(Hg)/HCl) and Wolff–Kishner (N_2H_4/KOH) give alkanes.
28	**A**	**S** — With the lowest-priority group horizontal, reverse the apparent direction: clockwise → S.
29	**D**	**Nucleophilic addition** — Nucleophile attacks the electrophilic carbonyl carbon.
30	**D**	**Less steric hindrance and fewer electron-donating alkyl groups** — Both steric and electronic reasons.
31	**B**	**sp^2 to sp^3** — A tetrahedral intermediate forms.
32	**B**	**A racemic mixture (attack from both faces)** — The planar carbonyl is attacked equally from both faces.
33	**B**	**Cyanohydrins** — Catalysed by base (CN^-).
34	**A**	**Acetals** — Hemiacetal is the intermediate.
35	**D**	**Protecting groups for carbonyls (stable in base)** — Removed by aqueous acid.
36	**B**	**Imines (Schiff bases)** — Secondary amines give enamines.
37	**D**	**Oximes** — Used for identification.
38	**B**	**Yellow/orange-red precipitate (2,4-DNP derivative)** — A general test for C=O.
39	**B**	**Primary alcohol** — Other aldehydes → 2°; ketones → 3°.
40	**A**	**Tertiary alcohol** — Two R groups already on carbonyl carbon.
41	**A**	**Primary alcohols** — Ketones → secondary alcohols.

42	B	**CH_2** — For acid-stable compounds.
43	C	**NH_2NH_2/KOH, heat** — For base-stable compounds; C=O → CH_2.
44	D	**An alkene** — Using a phosphorus ylide.
45	D	**Aldehydes** — Aldehydes are oxidized to carboxylates.
46	B	**Cu_2O** — Aliphatic aldehydes reduce Cu^{2+}.

Lecture 3

47	B	**Benzaldehyde** — Cannizzaro needs an aldehyde without α-hydrogens (benzaldehyde, formaldehyde).
48	B	**CH_3CO^-** — Methyl ketones (and compounds that give them) form yellow CHI_3; the product has one carbon fewer.
49	C	**An α-hydrogen and a strong base** — Base removes the α-H to make an enolate, which attacks a second carbonyl.
50	B	**4-chloropentanoic acid** — Five-carbon chain, COOH is C1, Cl on C4.
51	B	**Adjacent to the carbonyl** — It is acidic (pKa ~19–20 for ketones).
52	D	**Resonance-stabilized with charge on oxygen** — Delocalization onto O.
53	A	**Movement of a proton and a double bond** — Keto form usually predominates.
54	C	**At least one α-H** — Enolate attacks another carbonyl.
55	C	**3-Hydroxybutanal** — Dehydration gives crotonaldehyde (but-2-enal).
56	A	**An α,β-unsaturated carbonyl** — Loss of water.
57	C	**Lack α-H, in concentrated base** — E.g. HCHO, benzaldehyde.
58	A	**Benzyl alcohol + benzoate** — Disproportionation: one oxidized, one reduced.
59	A	**Oxidized to formate** — The other aldehyde is reduced.
60	B	**Methyl ketones and ethanol/2° alcohols $CH_3CH(OH)^-$** — Yellow CHI_3 precipitate.
61	A	**CHI_3** — Yellow solid with antiseptic smell.
62	B	**An aromatic aldehyde (no α-H) and a ketone/aldehyde with α-H** — A crossed aldol.
63	C	**C=O and –OH** — –COOH.
64	D	**Hydrogen-bonded dimers** — Higher than alcohols of similar mass.
65	C	**Ethanoic acid** — Common name acetic acid.
66	B	**Formic (methanoic) acid** — Found in ant stings.

Lecture 4

67	C	**FCH_2COOH** — F is the most electronegative, giving the strongest −I effect.
68	C	**The ortho effect (steric inhibition of resonance)** — The ortho group twists COOH out of the ring plane, removing destabilizing resonance.
69	A	**One more carbon** — CO_2 adds one carbon as the COOH group.
70	A	**Halogenates the α-carbon of acids (X_2/P)** — The α-halo acid can then become an α-hydroxy or α-amino acid.
71	D	**It is more water-soluble, so absorbed faster** — Higher solubility → faster absorption → quicker onset.
72	D	**Acid chloride** — Cl^- is a weak base and an excellent leaving group.
73	D	**Resonance-stabilized (two equivalent O)** — Charge shared between two oxygens.
74	C	**Increase acidity** — −I effect stabilizes the anion.
75	D	**Cl_3CCOOH** — More Cl, stronger −I effect.
76	A	**FCH_2COOH** — F is more electronegative.
77	A	**Distance from the carboxyl group** — 2-chlorobutanoic > 3-chloro > 4-chloro.
78	C	**CH_3 is electron-donating (+I), destabilizing acetate** — Alkyl groups reduce acidity.
79	D	**More acidic** — NO_2 withdraws by −I and −M.
80	B	**Less acidic** — OCH_3 donates by +M.
81	A	**Ortho effect (steric + electronic)** — Almost any ortho group increases acidity.
82	B	**CO_2** — Phenols generally do not — a distinguishing test.
83	C	**Benzoic acid** — Side-chain oxidation to –COOH.

84	**B**	**Carboxylic acid** — Via the amide.
85	**C**	**Carboxylic acid with one extra carbon** — Carboxylation.
86	**B**	**An alcohol in acid catalysis** — Reversible; water removed to drive it.
87	**C**	**Acid chlorides** — By-products SO_2 and HCl are gases.
88	**A**	**Primary alcohols** — $NaBH_4$ cannot reduce acids.
89	**B**	**Decarboxylation (loss of CO_2)** — Via a cyclic transition state.
90	**B**	**α-Halogen into carboxylic acids (Br_2/P)** — α-Bromination.

Lecture 5

91	**B**	**Chair** — Chair: 109.5° angles and all C–H bonds staggered.
92	**D**	**C1 and C4** — The two 'flagpole' H on C1 and C4 point toward each other (steric strain).
93	**D**	**An intramolecular hydrogen bond** — The two OH groups H-bond when 60° apart, outweighing the anti preference.
94	**C**	**Different spatial arrangements from rotation about single bonds** — Interconvert readily at room temperature.
95	**C**	**Staggered** — Less torsional strain.
96	**B**	**3 kcal/mol (12 kJ/mol)** — Small, so rotation is fast.
97	**A**	**Anti** — Methyls 180° apart.
98	**A**	**Steric strain between methyl groups at 60°** — ~0.9 kcal/mol.
99	**C**	**Chair** — Free of angle and torsional strain.
100	**B**	**109.5°** — Tetrahedral, no angle strain.
101	**D**	**Flagpole interactions and eclipsing** — Twist-boat relieves some strain.
102	**B**	**One axial and one equatorial H** — 6 axial + 6 equatorial positions.
103	**B**	**Parallel to the ring's axis (up/down)** — Equatorial bonds point outward around the ring.
104	**C**	**Equatorial** — And equatorial to axial.
105	**A**	**1,3-Diaxial interactions** — Steric clash with axial H on C3 and C5.
106	**D**	**Equatorial** — It 'locks' the conformation.
107	**A**	**One axial, one equatorial** — Trans-1,2 can be diequatorial.
108	**C**	**trans (diequatorial)** — Trans can be e,e.
109	**B**	**Diequatorial** — Cis-1,3 can be e,e.
110	**A**	**60° vs 109.5° ideal** — Baeyer angle strain.
111	**A**	**Along a C–C bond** — Front carbon a point, back carbon a circle.

Lecture 6

112	**C**	**Aziridine** — 3-membered + N: unsaturated -irine, saturated -iridine.
113	**C**	**59%** — About 59%, partly because N-heterocycles improve solubility and selectivity.
114	**A**	**At least one atom other than carbon** — Common heteroatoms: O, S, N.
115	**C**	**Oxa** — S = thia, N = aza.
116	**C**	**O > S > N** — O takes precedence and gets the lowest number.
117	**D**	**A vowel** — E.g. oxa + irane → oxirane.
118	**B**	**-irane** — With N: -iridine.
119	**B**	**-ole** — Saturated 5-ring: -olane (with N: -olidine).
120	**A**	**Oxolane** — Common name tetrahydrofuran.
121	**C**	**Oxole** — Unsaturated 5-ring with O.
122	**B**	**S** — Thiole in H–W naming.
123	**D**	**One NH** — Aromatic, lone pair in the π system.
124	**A**	**6-membered aromatic with one N** — Azine in H–W terms.
125	**B**	**Piperidine** — Saturated pyrrole = pyrrolidine.
126	**B**	**1,3** — Pyrazole is 1,2-diazole.

127	C	**1,2-Diazole** — Adjacent nitrogens.
128	B	**S and N at 1,3** — Present in vitamin B1 and penicillin.
129	B	**1,2** — Oxazole is 1,3.
130	B	**1,3** — Pyrazine is 1,4; pyridazine 1,2.
131	C	**1,4-Diazine** — Nitrogens para to each other.
132	A	**Pyrrole** — Benzopyrrole; in tryptophan.
133	A	**1** — Isoquinoline has N at 2.
134	D	**Pyrimidine and imidazole** — Adenine and guanine are purines.
135	B	**-epine** — E.g. azepine; 8-membered -ocine.
136	A	**The heteroatom of highest priority** — Then continues to give other heteroatoms lowest numbers.
137	A	**6 π electrons including a heteroatom lone pair** — Hückel 4n+2.